PROJECT PROPOSAL

Title: Google Play Store

1. Student Details

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2. Problem Statement

The Google Play Store dataset contains large amounts of information about mobile applications available on the Google Play Store, such as ratings, reviews, installs, and categories. However, it is difficult to directly understand patterns and trends from raw data.

The problem is to analyze and extract meaningful insights from this dataset to understand app performance, user preferences, and market trends. This includes identifying popular app categories, studying the relationship between ratings and installs, and comparing free and paid applications.

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3. Domain:

Education / Student Performance Analysis

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4. Dataset Information

- Dataset Name: Google Play Store.csv
- Source: Kaggle
- The Google Play Store dataset contains information about different mobile applications available on the Google Play Store. This dataset is used to analyze app performance, popularity, user ratings, and trends in the mobile application market.

- The dataset includes several attributes such as App name, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, and Last Updated date. These features help in understanding how different factors affect the success of an application.
- Using this dataset, data analysts can perform Exploratory Data Analysis (EDA) to identify patterns such as which app categories are most popular, how ratings affect installs, and how pricing impacts user downloads.
- In simple terms, the dataset helps researchers and developers understand user preferences, app popularity, and market trends in the Android application ecosystem.

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5. Dataset Description

- **Source:** Kaggle
- **Dataset Name:** Google Play Store
- **Dataset Link:** https://www.kaggle.com/datasets/lava18/google-play-store-apps

Dataset Features:

- **App – Name of the application**
- **Category – Type of app (Game, Education, Business, etc.)**
- **Rating – User rating (1 to 5)**
- **Reviews – Number of user reviews**
- **Size – Size of the app (MB/KB)**
- **Installs – Number of downloads/installs**
- **Type – Free or Paid app**
- **Price – Cost of the app**
- **Content Rating – Suitable age group (Everyone, Teen, etc.)**
- **Genres – App genre/category**
- **Last Updated – Date of last update**
- **Current Ver – Current version of the app**

 Android Ver – Required Android version

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6. Objectives

- To analyze app ratings and reviews to understand user satisfaction.
- To identify the most popular app categories based on installs.
- To study the relationship between installs, ratings, and reviews.
- To compare free and paid applications in terms of popularity.
- To understand trends in the mobile application market.

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7. Methodology

1. Data Collection

- The dataset is collected from online sources such as Kaggle.
- It contains information about apps like ratings, installs, reviews, and categories.
- The dataset helps analyze the performance and popularity of mobile applications.

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2. Data Loading

- The dataset is loaded into **Python** using the **Pandas** library.
- The CSV file is imported using pd.read_csv() function.
- The data is stored in a dataframe for further analysis.

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3. Data Cleaning

- Missing values in columns such as ratings or size are handled.
- Duplicate records are removed from the dataset.
- Data types are converted into the correct format for analysis.

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4. Exploratory Data Analysis (EDA)

- Basic statistics like mean, median, and standard deviation are calculated.
- Relationships between variables such as rating and installs are examined.
- Patterns and trends in app categories are identified.

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5. Data Visualization

- Graphs such as bar charts, scatter plots, and box plots are created.
- Visualization helps understand patterns and comparisons easily.
- Libraries like **Matplotlib** and **Seaborn** are used.

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6. Result Interpretation

- The results are analyzed to understand app popularity and user preferences.
- Insights about ratings, installs, and categories are identified.
- The findings help understand trends in the mobile app market.

5. Tools & Technologies

- **Python**
  Used for data analysis and processing.
- **Pandas**
  Used for data manipulation and handling datasets.
- **NumPy**
  Used for numerical calculations and operations.
- **Matplotlib**
  Used to create basic graphs and charts.
- **Seaborn**
  Used for advanced data visualization.
- **Google Colab** / Jupyter Notebook
  Used as the working platform to write and execute code.

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8. Expected Outcome

- Identify popular app categories.
- Understand app ratings and reviews.
- Compare free and paid apps.
- Analyze relationship between installs and ratings.
- Identify trends in the Google Play Store market.

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9. Conclusion

The analysis of the **Google Play Store** dataset helps understand the performance and popularity of mobile applications.
It shows how factors such as ratings, reviews, and installs influence the success of an app.
Data visualization and statistical analysis help identify trends in different app categories.
Overall, the study provides useful insights about user preferences and the mobile app market.